QUBO Formulation: Maximum Independent Set Problem

Maximum Independent Set Problem

Problem Statement. Let $G = (V, E)$ be an undirected graph with vertex set V and edge set E. An independent set is a subset $S \subseteq V$ such that no two vertices in S are connected by an edge. The Maximum Independent Set Problem seeks to find an independent set S of maximum cardinality, i.e., $\max |S|$ subject to the constraint that for every edge $(i, j) \in E$, at most one of i or j belongs to S.

Decision Variables. For each vertex $i \in V$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i takes the value 1 if vertex i is selected for the independent set, and 0 otherwise. The decision vector is denoted $\mathbf{x} = (x_0, x_1, \ldots, x_{n-1}) \in \{0, 1\}^n$, where $n = |V|$.

QUBO Derivation. Step 1 — Objective Function. The goal is to maximize the number of selected vertices, $\sum_{i \in V} x_i$. Since QUBO solvers minimize energy, we negate the objective:

$$\min_{\mathbf{x} \in \{0,1\}^n} \left(-\sum_{i \in V} x_i \right). \tag{1}$$

Step 2 — Constraint Formulation. For each edge $(i, j) \in E$, the independence condition requires that we cannot select both endpoints simultaneously. This is enforced by the linear inequality:

$$x_i + x_j \leq 1 \quad \forall (i, j) \in E. \tag{2}$$

Step 3 — Penalty Term Expansion. Each inequality is converted to a quadratic penalty term by squaring the violation. For a given edge $(i, j) \in E$ and penalty weight $\lambda > 0$, the penalty is:

$$P_{(i,j)}(\mathbf{x}) = \lambda (x_i + x_j - 1)^2 \tag{3}$$

$$= \lambda (x_i^2 + x_j^2 + 1 + 2x_i x_j - 2x_i - 2x_j). \tag{4}$$

Using the idempotent property of binary variables, $x_k^2 = x_k$ for all $k \in V$, the expression simplifies to:

$$P_{(i,j)}(\mathbf{x}) = \lambda(x_i + x_j + 1 + 2x_ix_j - 2x_i - 2x_j) \tag{5}$$
$$= \lambda(1 - x_i - x_j + 2x_ix_j). \tag{6}$$

This is the general algebraic contribution of edge (i, j) to the Hamiltonian.

Step 4 — Combined Hamiltonian. Summing the negated objective and all edge penalties yields the total QUBO energy function $H(\mathbf{x})$:

$$H(\mathbf{x}) = -\sum_{i \in V} x_i + \sum_{(i,j) \in E} \lambda(1 - x_i - x_j + 2x_ix_j). \tag{7}$$

Step 5 — Q Matrix and Offset. Collecting coefficients for linear and quadratic terms, and noting that each vertex i appears in $\deg(i)$ edges, we obtain the closed-form QUBO parameters:

$$Q_{i,i} = -1 - \lambda \deg(i) \quad \forall i \in V, \tag{8}$$
$$Q_{i,j} = 2\lambda \quad \forall (i, j) \in E, \tag{9}$$
$$Q_{i,j} = 0 \quad \forall (i, j) \notin E, \tag{10}$$
$$c = \lambda|E|. \tag{11}$$

The Hamiltonian is thus expressed as $H(\mathbf{x}) = \sum_{i \in V} Q_{i,i}x_i + \sum_{i<j} Q_{i,j}x_ix_j + c$.

Penalty Weight Justification. The penalty weight λ must be chosen large enough to ensure that any violation of the independence constraint increases the total energy relative to the best feasible assignment. Violating a single edge constraint by setting $x_i = x_j = 1$ instead of the optimal valid configuration $(1, 0)$ or $(0, 1)$ yields a maximum objective gain of 1 (selecting one extra vertex). The penalty incurred for this violation is $\lambda(1 + 1 - 1)^2 = \lambda$. Therefore, any $\lambda > 1$ strictly penalizes infeasible configurations more than the objective function rewards them. We select $\lambda = 2$ to provide a robust margin above the theoretical lower bound.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0, 1\}^n$ recovers the optimal maximum independent set because (i) any feasible independent set satisfies $x_i + x_j \leq 1$ for all $(i, j) \in E$, causing all penalty terms to vanish and reducing $H(\mathbf{x})$ to $-\sum_{i \in V} x_i$. Thus, among feasible solutions, minimizing H is exactly equivalent to maximizing the set size. (ii) Any infeasible assignment violates at least one edge constraint, incurring a penalty of at least λ. Since $\lambda > 1$, the energy of any infeasible solution strictly exceeds that of the best feasible solution on the violated edge(s). Consequently, the global minimum of H must be feasible and correspond to the maximum independent set.

Illustrative Example. Consider a path graph with $n = 3$ vertices $V = \{0, 1, 2\}$ and edges $E = \{(0, 1), (1, 2)\}$. The degrees are $\deg(0) = 1$, $\deg(1) = 2$, and $\deg(2) = 1$. Using $\lambda = 2$, the general formulas from Step 5 yield the concrete QUBO parameters:

$$Q_{0,0} = -1 - 2(1) = -3, \tag{12}$$
$$Q_{1,1} = -1 - 2(2) = -5, \tag{13}$$
$$Q_{2,2} = -1 - 2(1) = -3, \tag{14}$$
$$Q_{0,1} = 2(2) = 4, \tag{15}$$
$$Q_{1,2} = 2(2) = 4, \tag{16}$$
$$c = 2(2) = 4. \tag{17}$$

The resulting Hamiltonian is:

$$H(\mathbf{x}) = 4 - 3x_0 - 5x_1 - 3x_2 + 4x_0x_1 + 4x_1x_2. \tag{18}$$

Evaluating H over all $2^3 = 8$ binary assignments, the feasible independent sets are $(1, 0, 1)$, $(1, 0, 0)$, and $(0, 0, 1)$. Their energies are:

$$H(1, 0, 1) = 4 - 3 - 0 - 3 + 0 + 0 = -2, \tag{19}$$
$$H(1, 0, 0) = 4 - 3 - 0 - 0 + 0 + 0 = 1, \tag{20}$$
$$H(0, 0, 1) = 4 - 0 - 0 - 3 + 0 + 0 = 1. \tag{21}$$

Infeasible assignments such as $(1, 1, 0)$ incur a penalty of $2(1 + 1 - 1)^2 = 2$, yielding $H(1, 1, 0) = -2 + 2 = 0$. The global minimum is $H^* = -2$ at $\mathbf{x}^* = (1, 0, 1)$, correctly identifying the maximum independent set $\{0, 2\}$ of size 2.